

① Using Taylor Series, compute the following limits:

a) $\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5}$

Let $f(x) = \sin x$

$$\sin x = T_5(x) + R_5(x) = x - \frac{x^3}{3!} + \frac{x^5}{5!} + \frac{f^{(6)}(c)}{6!} x^6$$

Since $f^{(6)}(t) = \sin t \Rightarrow \sin x = x - \frac{x^3}{6} + \frac{x^5}{120} + \frac{\sin(c)}{720} x^6$

$$\sin x - x + \frac{x^3}{6} = \frac{x^5}{120} + \frac{\sin(c)}{720} x^6 \quad | : x^5$$

$$\frac{\sin x - x + \frac{x^3}{6}}{x^5} = \frac{1}{120} + \frac{\sin(c)}{720} \cdot x$$

$c \in (0, x) \Rightarrow$ as $x \rightarrow 0$, $c \rightarrow 0 \Rightarrow \sin(c)$ stays bounded

$$\lim_{x \rightarrow 0} \frac{\sin x - x + \frac{x^3}{6}}{x^5} = \lim_{x \rightarrow 0} \left(\frac{1}{120} + \underbrace{\sin(c)}_{\text{bounded}} \cdot \underbrace{\frac{x}{720}}_0 \right) = \frac{1}{120}$$

b) $\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x - \frac{3x^2}{2}}{x^4}$

Let $g(t) = e^t$

For $t = x^2$, $\exists c_1$ between 0 and x^2 s.t.:

$$\begin{aligned} e^{x^2} &= 1 + x + \frac{x^4}{2} + \frac{g^{(3)}(c_1)}{3!} (x^2)^3 = \\ &= 1 + x^2 + \frac{x^4}{2} + \frac{e^{c_1}}{6} \cdot x^6 \end{aligned}$$

Let $h(x) = \cos x \Rightarrow \exists c_2$ between 0 and x s.t.

$$\cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} + \frac{h^{(5)}(c_2)}{5!} x^5$$

$(h^{(5)}(t) = -\sin t)$

$$\Rightarrow \cos x = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{\sin(c_2)}{120} x^5$$

$$\begin{aligned}
 & e^{x^2} - \cos x - \frac{3x^2}{2} = \\
 & = \left(1 + x^2 + \frac{x^4}{2} + \frac{e^{c_1}}{6} x^6\right) - \left(1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{\sin(c_2)}{120} x^5\right) - \frac{3x^2}{2} = \\
 & = \left(x^2 + \frac{x^2}{2} - \frac{3x^2}{2}\right) + \left(\frac{x^4}{2} - \frac{x^4}{24}\right) + \frac{\sin(c_2)}{120} x^5 + \frac{e^{c_1}}{6} x^6 = \\
 & = 0 + \frac{11}{24} x^4 + \frac{\sin(c_2)}{120} x^5 + \frac{e^{c_1}}{6} x^6
 \end{aligned}$$

$$e^{x^2} - \cos x - \frac{3x^2}{2} = \frac{11}{24} x^4 + \frac{\sin(c_2)}{120} x^5 + \frac{e^{c_1}}{6} x^6 \quad / : x^4$$

$$\frac{e^{x^2} - \cos x - \frac{3x^2}{2}}{x^4} = \frac{11}{24} + \frac{\sin(c_2)}{120} x + \frac{e^{c_1}}{6} x^2$$

c_1 is between 0 and $x^2 \Rightarrow$ as $x \rightarrow 0$, $c_1 \rightarrow 0$

c_2 is between 0 and $x \Rightarrow$ as $x \rightarrow 0$, $c_2 \rightarrow 0$

Also $\Rightarrow \sin(c_2)$ stays bounded and $e^{c_1} \rightarrow 1$

$$\lim_{x \rightarrow 0} \frac{e^{x^2} - \cos x - \frac{3x^2}{2}}{x^4} = \lim_{x \rightarrow 0} \left(\frac{11}{24} + \underbrace{\sin(c_2)}_{\text{bounded}} \cdot \underbrace{\frac{x}{120}}_{\rightarrow 0} + \underbrace{\frac{e^{c_1}}{6}}_{\rightarrow \frac{1}{6}} \cdot \underbrace{\frac{x^2}{6}}_{\rightarrow 0} \right) = \frac{11}{24}$$

② Prove that the Taylor Series of $\ln(1+x)$ around 0 is

$$\ln(1+x) = \sum_{n \geq 1} (-1)^{n+1} \cdot \frac{x^n}{n}$$

$$\text{Let } f(x) = \ln(1+x)$$

$$f'(x) = \frac{1}{1+x} = (1+x)^{-1}$$

$$f''(x) = -\frac{1}{(1+x)^2} = -(1+x)^{-2}$$

$$f'''(x) = 2(1+x)^{-3}$$

$$f^{(4)}(x) = -6(1+x)^{-4}$$

$$\Rightarrow f^{(n)}(x) = (-1)^{n-1} \cdot (n-1)!$$

The Taylor series is:

$$f(x) = \sum_{n \geq 0} \frac{f^{(n)}(0)}{n!} \cdot x^n$$

For $n=0$, $f(0)=0 \Rightarrow$ constant term 0

For $n \geq 1$

$$\frac{f^{(n)}(0)}{n!} = \frac{(-1)^{n-1} (n-1)!}{n!} = \frac{(-1)^{n-1}}{n}$$

$$\Rightarrow \ln(1+x) = \sum_{n \geq 1} \frac{(-1)^{n-1}}{n} \cdot x^n$$

$$(-1)^{n-1} = (-1)^{n+1}$$

$$\Rightarrow \ln(1+x) = \sum_{n \geq 1} (-1)^{n+1} \cdot \frac{x^n}{n}$$

③ Using Taylor Series, prove that the forward difference $(f(x+h) - f(x))/h$ approximates the derivative $f'(x)$ with an error of order h and that the centered difference $(f(x+h) - f(x-h))/2h$ approximates the derivative $f'(x)$ with an error of order h^2 .

Assume f is at least 3 times differentiable for the forward difference and at least 4 times for the centered difference, on an interval containing x .

• Forward difference: Apply Lagrange remainder to $f(x+h)$, about x with $m=1$

$$f(x+h) = f(x) + h \cdot f'(x) + \frac{h^2}{2} f''(x + c_1 h), \text{ for } c_1 \in (0,1)$$

$$f(x+h) - f(x) = h \cdot f'(x) + \frac{h^2}{2} f''(x + c_1 h) \quad | : h$$

$$\frac{f(x+h) - f(x)}{h} = f'(x) + \frac{h}{2} f''(x + c_1 h)$$

$$f'(x) = \frac{f(x+h) - f(x)}{h} - \frac{h}{2} f''(x + c_1 h)$$

f'' is continuous near $x \Rightarrow$ remainder term $= \frac{h}{2} f''(x + c_1 h)$

is proportional to h $\xrightarrow[\text{Notation}]{\text{Big Oh}}$ $f'(x) = \frac{f(x+h) - f(x)}{h} + O(h)$

- Centered difference: Lagrange remainder for $f(x+h)$ and $f(x-h)$, about x , with $m=2$

$$\begin{cases} f(x+h) = f(x) + hf'(x) + \frac{h^2}{2} f''(x) + \frac{h^3}{6} f^{(3)}(x+c_2 h), \text{ for } c_2 \in (0,1) \\ f(x-h) = f(x) - hf'(x) + \frac{h^2}{2} f''(x) - \frac{h^3}{6} f^{(3)}(x+c_3(-h)), \text{ for } c_3 \in (0,1) \end{cases}$$

⊖

$$f(x+h) - f(x-h) = 2hf'(x) + \frac{h^3}{6} (f^{(3)}(x+c_2 h) + \underbrace{f^{(3)}(x+c_3(-h))}_{\text{write as } f^{(3)}(x-c_4 h), c_4 \in (0,1)}) \quad / : 2h$$

$$\frac{f(x+h) - f(x-h)}{2h} = f'(x) + \frac{h^2}{12} (f^{(3)}(x+c_2 h) + f^{(3)}(x-c_4 h))$$

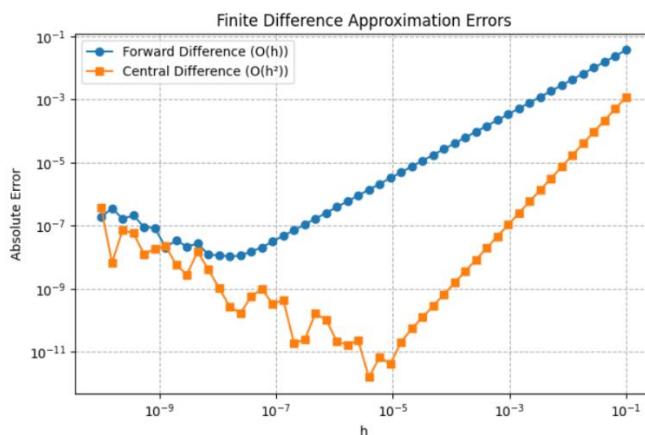
$\xrightarrow[\text{Notation}]{\text{Big O h}}$

$$f'(x) = \frac{f(x+h) - f(x-h)}{2h} + O(h^2)$$

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1 import numpy as np
2 import matplotlib.pyplot as plt
3
4 # choose a functions
5 def f(x): 4 usages
6     return np.sin(x)
7
8 # find its exact derivative
9 # (sinx)'= cosx
10 def f_prime_exact(x): 1 usage
11     return np.cos(x)
12
13 # point of interest = pi/4
14 x0 = np.pi / 4
15 f_prime_exact_val = f_prime_exact(x0)
16
17 # range of small h values
18 h_values = np.logspace(-10, -1, num=50) # 50 points from 1e-10 to 0.1
19
20 # compute finite difference approximations
21 forward_diff = (f(x0 + h_values) - f(x0)) / h_values
22 central_diff = (f(x0 + h_values) - f(x0 - h_values)) / (2 * h_values)
23
24 # compute absolute errors
25 error_forward = np.abs(forward_diff - f_prime_exact_val)
26 error_central = np.abs(central_diff - f_prime_exact_val)
27
28 # plot errors
29 plt.figure(figsize=(8,5))
30 plt.loglog(*args: h_values, error_forward, 'o-', label='Forward Difference (O(h))')
31 plt.loglog(*args: h_values, error_central, 's-', label='Central Difference (O(h^2))')
32 plt.xlabel('h')
33 plt.ylabel('Absolute Error')
34 plt.title("Finite Difference Approximation Errors")
35 plt.grid(visible=True, which="both", ls="--")
36 plt.legend()
37 plt.show()

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When h is very small, subtracting $f(x + h) - f(x)$ involves nearly equal numbers. This causes rounding errors in the computer's finite-precision arithmetic. Dividing by tiny h then amplifies these errors, so instead of decreasing, the total error starts increasing.